

# Dr. Sushant Kumar

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**Department of Electronics and Communication Engineering,**  
**Indian Institute of Information Technology (IIIT) Pune,**  
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**Google Scholar:** <https://scholar.google.com/citations?user=lwTkgu8AAAAJ&hl=en>



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## Research Interest:

Signal processing for wireless communication, synchronization parameters estimation, modulation classification, MIMO, OFDM, MIMO-OFDM, SC-FDMA, Massive MIMO.

## Education:

**Ph.D.** 2015–2020  
Electrical Engineering, IIT Patna

**M. Tech** 2013–2015  
Specialization: Communication System Engineering  
Electrical Engineering, IIT Patna

## Ph.D. Thesis:

- **Title:** *Blind synchronization for single carrier, SC-FDMA, and MIMO SC-FDMA systems*
- **Supervisor:** Dr. Sudhan Majhi, Associate Professor, Dept of EE, IIT Patna.

## M.Tech Project:

- **Title:** *STO estimation for OQPSK modulated signal*
- **Supervisor:** Dr. Sudhan Majhi, Associate Professor, Dept of EE, IIT Patna.

## Experience:

- **Assistant Professor, IIIT, Pune** Dec 2021 to Present
- **Assistant Professor (Contract), NIT, Patna** Dec 2020 to Dec 2021
- **Research Associate, IIT Patna** August 2020 to Dec 2020
- **Teaching Assistant, IIT Patna** 2016-2019

## Journal Publications:

1. M. S. Chaudhari, S. Kumar, R. Gupta, M. Kumar, and S. Majhi, "Implementation and measurement of blind receiver for OFDM systems", *IEEE Transactions on Instrumentation and Measurements*, November 2021.  
(Impact factor: 4.016), DOI: [10.1109/TIM.2021.3124833](https://doi.org/10.1109/TIM.2021.3124833).
2. K. Pathy, A. Kumar, R. Gupta, S. Kumar, and S. Majhi, "Design and implementation of blind modulation classification for asynchronous MIMO-OFDM system", *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1-11, 2021, Art no. 5504011.  
(Impact factor: 4.016), DOI: [10.1109/TIM.2021.3109737](https://doi.org/10.1109/TIM.2021.3109737).

3. S. Kumar, M. S. Chaudhari, R. Gupta, and S. Majhi, "Multiple CFOs Estimation and Implementation of SC-FDMA Uplink System Using Oversampling and Iterative Method", *IEEE Transactions on Vehicular Technology*, vol. 69, no. 6, pp. 6254-6263, June 2020. (Impact factor: 5. 978), DOI: [10.1109/TVT.2020.2984059](https://doi.org/10.1109/TVT.2020.2984059).
4. R. Gupta, S. Kumar, and S. Majhi, "Blind modulation classification for asynchronous OFDM systems over unknown signal parameters and channel statistics", *IEEE Transactions on Vehicular Technology*, vol. 69, no. 5, pp. 5281-5292, May 2020. (Impact factor: 5.978), DOI: [10.1109/TVT.2020.2981935](https://doi.org/10.1109/TVT.2020.2981935).
5. S. Kumar, S. Majhi and C. Yuen, "Multi-user CFOs estimation for SC-FDMA system over frequency selective fading channels," *IEEE Access*, vol. 6, pp. 43146-43156, Aug. 2018. (Impact factor: 3.745), DOI: [10.1109/ACCESS.2018.2856578](https://doi.org/10.1109/ACCESS.2018.2856578).
6. S. Kumar, and S. Majhi, "Blind symbol timing offset estimation for offset-QPSK modulated signals," *Wiley ETRI Journal*, vol. 42, no. 3, pp. 324-332, November 2019. (Impact factor: 1.347), DOI: [10.4218/etrij.2019-0199](https://doi.org/10.4218/etrij.2019-0199).

#### **Manuscripts under review**

1. S. Kumar, and S. Majhi, "Design and testbed implementation of multiuser CFOs estimation for MIMO SC-FDMA system", *communicated to IEEE Transactions on Signal Processing*.

#### **Conference Publications/Presentations:**

1. R. Gupta, S. Kumar, and S. Majhi, "Blind modulation classification for OFDM in the presence of timing, frequency, and phase offsets", in Proc. **90th IEEE VTC Fall**, 2019.
2. Harsha, S. Kumar, and S. Majhi, "Blind CFO Estimation for Multi-user in SC-FDMA Uplink Systems Using Variance Minimization", in Proc **17th IEEE IWCMC 2021**, Harbin, China, pp. 1971-1976. DOI: 10.1109/IWCMC51323.2021.9498739.
3. Harsha, S. Kumar, S Singh and S. Majhi, "CFO Estimation for Multi-user Uplink SC-FDMA Using Null Subcarrier and Deterministic Approach", accepted in **4th IEEE International Conference on Information and Communications Technology (ICOIACT) 2021**. DOI: [10.1109/ICOIACT53268.2021.9563970](https://doi.org/10.1109/ICOIACT53268.2021.9563970)

#### **Awards/Achievements:**

- Qualified GATE in 2013 with all India rank 800.
- Qualified UGC-NET June-2015 in electronics Science.
- Obtained 100% in Mathematics in class 10, CBSE.
- Merit Fellowship (2007-2011) during B. Tech at ICFAI University.

#### **International Travel Grant:**

- Received travel grant from MHRD, New Delhi India, to attend the 90th IEEE Vehicular Technology Conference (VTC2019-Fall) during 22-25 September 2019 at Honolulu, Hawaii, USA.

#### **Professional Services: (Publons Profile)**

- IEEE Transactions on Vehicular Technology
- IEEE Communications letters
- SPRINGER: Circuits, Systems, and Signal Processing